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Discrete–time Field Oriented Control for Induction Motors

Iván E. Dueñas, * **Jorge Rivera,** ** **Stefano Di Gennaro,** ***,**** **Susana Ortega** *

* *Electronic design department, CINVESTAV, Jalisco, Mexico, (e–mail ieduenas@gdl.cinvestav.mx sortega@gdl.cinvestav.mx*
** *CONACyT–CINVESTAV, Jalisco, Mexico (e–mail: jriverado@conacyt.mx)*
*** *Department of Information Engineering, Computer Science and Mathematics Center of Excellence DEWS University of L'Aquila, L'Aquila, Italy (e–mail: stefano.digennaro@univaq.it)*
**** *Center of Excellence DEWS, University of L'Aquila, L'Aquila, Italy*

Abstract: In this work it is proposed a discrete–time field oriented control for induction motors. From a discrete–time stationary reference frame model for an induction motor, obtained by a variational integrator, the model is converted to a field oriented frame. Then a discrete–time field oriented control is designed for the asymptotic regulation of the rotor velocity and the rotor flux. Simulations are carried out for the validation of the proposed control scheme.

Keywords: Field oriented control; Induction motors; Discrete–time.

1. INTRODUCTION

Due to the easy maintenance and low cost, the popularity of three phase Induction Motors (IMs) has risen considerably. The IM is a nonlinear dynamic system with time–variant parameters. This makes the design of a control law a challenging task (Krause et al. (2002)). Field Oriented Control (FOC) technique is mostly used on this type of motors because it allows simplifying and decomposing the control design, similarly to what can be done for a DC motor with field winding (Blaschke (1972)).

Digital devices are commonly used for implementing control algorithms, such as digital signal processors (DSP) and field programmable gate arrays (FPGAs) (Monmasson and Cirstea (2007)). Usually, for these devices the control algorithms are designed in continuous–time, and implemented digitally. This implies that the control algorithms have to be approximated in discrete–time. A drawback of this method is that the performance of the closed–loop system may be reduced, and the control gains must be fine–tuned. To overcome this problem, a common approach is the use of the explicit Euler method for the discrete–time approximation of the continuous–time control. As a result, high sampling rates are required for the implementation of such algorithms, requiring faster digital devices. This clearly increases the total power consumption and cost of the digital implementation. An alternative solution is to obtain a discrete–time model of the IM, and to use it to design a discrete–time controller (Castillo-Toledo et al. (2008)). Clearly, the quality of the solution will depend on the accuracy of the discrete–time model, approximating the sampled dynamics of the IM.

With the aim of designing discrete–time FOC controllers, in Ortega and Taoutaou (1996) an exact discrete–time model of a current–fed (reduced order model) IM was presented, and used for designing a discrete–time FOC controller. Based on a discrete–time model obtained by means of the explicit Euler method, discrete–time FOC controllers were designed in Neves et al. (1999), Veselic et al. (2010), where high sampling rates are required. Moreover, in Veselic et al. (2010), Gao's discrete–time reaching law was used, which employs a sign function causing chattering problems. In Proca et al. (2002), the reference stator currents were designed in continuous time, whereas the stator current dynamics were approximated in discrete–time for a FOC law design. A sensorless discrete–time FOC was designed in Vieira et al. (2012), where the explicit Euler method was used under the assumption of constant rotor velocity, and a sign function was used in the control design.

To the best of the authors' knowledge, there is not a full–order discrete–time model oriented to the field that excels the performance of the model obtained by means of the explicit Euler method. But in the work presented by Rivera (2015a), a novel discrete–time model for IM in the stationary (α, β) reference frame was obtained, using a variational integrator technique. The resulting model approximates better, at high sampling times, the sampled dynamics of an IM than the model obtained by the explicit Euler method.

Therefore, based on the discrete–time model for IM presented by Rivera (2015a), in this work, such model is oriented to the field, i.e. to the reference frame that rotates

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with rotor flux angle (the (d,q) reference frame). Then, a discrete–time FOC is designed based on classic PI control actions.

The rest of this work is organized as follows. In Section 2 the discrete–time transformation of the IM to the frame oriented to the field is presented. In Section 3, the discrete–time FOC is designed. Simulation results are presented in Section 4. Finally, some conclusions are presented in Section 5.

2. DISCRETE–TIME FIELD ORIENTATION FOR IMS

The IM model presented in Rivera (2015a) corresponds to the (α,β) model in the stationary reference frame

$$\begin{aligned}
\omega_{k+1} &= \omega_k - \frac{f_v}{J}h\omega_k + \mu_d h I^T_{\alpha,\beta,k}\mathcal{J}\Phi_{\alpha,\beta,k} - \frac{h}{J}\tau_{L,k}\\
\Phi_{\alpha,\beta,k+1} &= \alpha_d e^{\mathcal{J}hp\omega_k}\Phi_{\alpha,\beta,k} + L_m\alpha\alpha_d h e^{\mathcal{J}hp\omega_k} I_{\alpha,\beta,k}\\
I_{\alpha,\beta,k+1} &= \gamma_d I_{\alpha,\beta,k} + \beta_d\alpha_d\Phi_{\alpha,\beta,k} - \beta_d\alpha_d^2 e^{\mathcal{J}hp\omega_k}\\
&\quad\times(\Phi_{\alpha,\beta,k} + L_m\alpha h I_{\alpha,\beta,k}) + \frac{h}{\sigma_d}U_{\alpha,\beta,k}
\end{aligned}\tag{1}$$

where ω_k is the rotor velocity, $\tau_{L,k}$ is the load torque, $\Phi_{\alpha,\beta,k} = (\phi_{\alpha,k},\phi_{\beta,k})^T$ is the rotor flux vector, $I_{\alpha,\beta,k} = (i_{\alpha,k}, i_{\beta,k})^T$ is the stator current vector, $U_{\alpha,\beta,k} = (u_{\alpha,k}, u_{\beta,k})^T$ is the input voltage vector,

$$\mathcal{J} = \begin{pmatrix} 0 & -1\\ 1 & 0\end{pmatrix}$$

is a skew–symmetric matrix, and

$$e^{\mathcal{J}hp\omega_k} = \begin{pmatrix}\cos hp\omega_k & -\sin hp\omega_k\\ \sin hp\omega_k & \cos hp\omega_k\end{pmatrix}$$

with h the sampling period. Moreover, $\mu_d = \mu\alpha_d$, $\alpha_d = 1/(1+\alpha h)$, $\gamma_d = 1 - R_s h/\sigma_d$, $\beta_d = L_m/(L_r\sigma_d)$, $\sigma_d = \sigma + \alpha\alpha_d^2 h L_m^2/L_r$, $\mu = 3pL_m/(2JL_r)$, $\alpha = R_r/L_r$, $\sigma = L_s - L_m^2/L_r$, where the parameters of the IM are R_s, R_r as the stator, rotor resistances, L_s, L_r, L_m as the stator, rotor, mutual inductances, J as the inertia moment, p as the number of poles, and f_v as the mechanical viscous friction coefficient.

Considering the rotation matrix (Krause et al. (2002))

$$e^{-\mathcal{J}\theta_{\phi,k}} = \begin{pmatrix}\cos\theta_{\phi,k} & \sin\theta_{\phi,k}\\ -\sin\theta_{\phi,k} & \cos\theta_{\phi,k}\end{pmatrix}\tag{2}$$

with

$$\theta_{\phi,k} = \arctan\left(\frac{\phi_{\beta,k}}{\phi_{\alpha,k}}\right)$$

the IM model (1) in the stationary reference frame can be transformed into the frame rotating with the rotor flux vector, obtaining

$$f_{d,q,k} = e^{-\mathcal{J}\theta_{\phi,k}} f_{\alpha,\beta,k}$$

where f represents Φ, I, or U. The resulting discrete–time model for IM oriented to the field is

$$\begin{aligned}
\omega_{k+1} &= \omega_k - \frac{f_v}{J}h\omega_k + \mu_d h i_{q,k}\phi_{d,k} - \frac{h}{J}\tau_{L,k}\\
\Phi_{d,0,k+1} &= \alpha_d e^{-\mathcal{J}h\omega_{s,k}}\Phi_{d,0,k} + L_m\alpha\alpha_d h e^{-\mathcal{J}h\omega_{s,k}} I_{d,q,k}\\
I_{d,q,k+1} &= \gamma_d e^{-\mathcal{J}h\omega_{\phi,k}} I_{d,q,k} + \beta_d\alpha_d e^{-\mathcal{J}h\omega_{\phi,k}}\Phi_{d,0,k}\\
&\quad - \beta_d\alpha_d^2 e^{-\mathcal{J}h\omega_{s,k}}(\Phi_{d,0,k} + L_m\alpha h I_{d,q,k})\\
&\quad + \frac{h}{\sigma_d}e^{-\mathcal{J}h\omega_{\phi,k}} U_{d,q,k}
\end{aligned}\tag{3}$$

where $\omega_{s,k} = \omega_{\phi,k} - p\omega_k$ is the slip angular velocity, $\omega_{\phi,k}$ is the synchronous velocity of the rotor flux, and $\Phi_{d,0,k} = (\phi_{d,k}, 0)^T$.

3. DISCRETE–TIME FIELD ORIENTED CONTROL

A direct discrete–time FOC will be designed, using two PI control loops applied to the discrete–time model (3) oriented to the field. The rotor flux vector and the load torque are estimated with observers, designed in the stationary reference frame using ω_k and $I_{d,q,k}$.

3.1 Control design

Let us define the output tracking error

$$z_{1,k} = y_k - y_{r,k}\tag{4}$$

where $z_{1,k} = (z_{11,k}, z_{12,k})^T = (\omega_k, \phi_{d,k})^T$, and $y_{r,k} = (\omega_{r,k}, \phi_{d,r,k})^T$. Taking one step ahead, one gets the dynamics

$$z_{1,k+1} = f_{1,k} + g_{1,k} I_{d,q,k}$$

where

$$f_{1,k} = \begin{pmatrix}\omega_k - \frac{f_v}{J}h\omega_k - \frac{h}{J}\tau_{L,k} - \omega_{r,k+1}\\ \alpha_d\phi_{d,k}\cos(h\omega_{s,k}) - \phi_{d,r,k+1}\end{pmatrix}$$

$$g_{1,k} = \begin{pmatrix} 0 & \mu_d h\phi_{d,k}\\ L_m\alpha\alpha_d h\cos(h\omega_{s,k}) & L_m\alpha\alpha_d h\sin(h\omega_{s,k})\end{pmatrix}.$$

The desired dynamics can be assigned imposing

$$z_{1,k+1} = f_{1,k} + g_{1,k}I_{d,q,k} = K_{1,p}z_{1,k} + K_{1,i}z_{i,1,k}\tag{5}$$

with $K_{1,p} = \mathrm{diag}(k_{11,p}, k_{12,p})$, $K_{1,i} = \mathrm{diag}(k_{11,i}, k_{12,i})$, and $z_{i,1,k}$ a discrete–time integral action, defined as

$$z_{i,1,k+1} = z_{i,1,k} + hz_{1,k}.$$

Hence, the desired stator currents are solved from (5), and defined as reference signals

$$I_{d,q,r,k} = g_{1,k}^{-1}(K_{1,p}z_{1,k} + K_{1,i}z_{i,1,k} - f_{1,k})\tag{6}$$

with $I_{d,q,r,k} = (i_{d,r,k}, i_{q,r,k})^T$. Then, a second error is defined for the tracking of the stator reference currents

$$z_{2,k} = I_{d,q,k} - I_{d,q,r,k}\tag{7}$$

with $z_{2,k} = (z_{21,k}, z_{22,k})^T$. Again, taking one step ahead, the following difference equation is obtained

$$z_{2,k+1} = f_{2,k} + g_{2,k}U_{d,q,k}$$

with

$$\begin{aligned}
f_{2,k} &= \gamma_d e^{-\mathcal{J}h\omega_{\phi,k}} I_{d,q,k} + \beta_d\alpha_d e^{-\mathcal{J}h\omega_{\phi,k}}\Phi_{d,0,k}\\
&\quad - \beta_d\alpha_d^2 e^{-\mathcal{J}h\omega_{s,k}}(\Phi_{d,0,k} + L_m\alpha h I_{d,q,k}) - I_{d,q,r,k+1}\\
g_{2,k} &= \frac{h}{\sigma_d}e^{-\mathcal{J}h\omega_{\phi,k}}.
\end{aligned}$$

The control input $U_{d,q,k}$ is then selected for cancelling the old dynamics, and for introducing the new ones

$$U_{d,q,k} = g_{2,k}^{-1}(K_{2,p}z_{2,k} + K_{2,i}z_{i,2,k} - f_{2,k})\tag{8}$$

with $K_{2,p} = \mathrm{diag}(k_{21,p}, k_{22,p})$, $K_{2,i} = \mathrm{diag}(k_{21,i}, k_{22,i})$, and $z_{i,2,k}$ a discrete–time integral action defined as

$$z_{i,2,k+1} = z_{i,2,k} + hz_{2,k}.$$

The transformations (4) and (7) define a diffeomorphism for (3), yielding the following closed–loop system

$$\begin{aligned}
z_{1,k+1} &= K_{1,p}z_{1,k} + K_{1,i}z_{i,1,k} + g_{1,k}z_{2,k}\\
z_{i,1,k+1} &= z_{i,1,k} + hz_{1,k}\\
z_{2,k+1} &= K_{2,p}z_{2,k} + K_{2,i}z_{i,2,k}\\
z_{i,2,k+1} &= z_{i,2,k} + hz_{2,k}.
\end{aligned}$$

Choosing $K_{2,p}$ as a Schur matrix and $K_{2,i}$ as a negative definite matrix, one has

$$\lim_{k\to\infty} z_{2,k} = 0$$

and the closed–loop system reduces to the following

$$z_{1,k+1} = K_{1,p} z_{1,k} + K_{1,i} z_{i,1,k}$$
$$z_{i,1,k+1} = z_{i,1,k} + h z_{1,k}.$$

Similarly, if $K_{1,p}$ is chosen as a Schur matrix and $K_{1,i}$ as a negative definite matrix, one gets

$$\lim_{k\to\infty} z_{1,k} = 0.$$

3.2 Rotor flux observer

The flux vector is very difficult to measure. A super twisting sliding mode flux observer can be used to estimate such a flux, along with the current, as in Rivera (2015b)

$$\begin{aligned}\hat{\Phi}_{\alpha,\beta,k+1} &= \alpha_d e^{\mathcal{J}hp\omega_k}\hat{\Phi}_{\alpha,\beta,k} + L_m\alpha\alpha_d h e^{\mathcal{J}hp\omega_k} I_{\alpha,\beta,k} \\ &\quad - k_v V_k \\ \hat{I}_{\alpha,\beta,k+1} &= \gamma_d \hat{I}_{\alpha,\beta,k} + \beta_d\alpha_d\hat{\Phi}_{\alpha,\beta,k} - \beta_d\alpha_d^2 e^{\mathcal{J}hp\omega_k} \\ &\quad \times (\hat{\Phi}_{\alpha,\beta,k} + L_m\alpha h I_{\alpha,\beta,k}) + \frac{h}{\sigma_d}U_{\alpha,\beta,k} - V_k\end{aligned}$$

with $\hat{\Phi}_{\alpha,\beta,k}$, $\hat{I}_{\alpha,\beta,k}$ the estimates of the rotor flux vector and of the stator current vector, respectively, and V_k the signal to be injected in the observer.

The estimation errors are defined as

$$\tilde{\Phi}_{\alpha,\beta,k} = \Phi_{\alpha,\beta,k} - \hat{\Phi}_{\alpha,\beta,k} \tag{9}$$
$$\tilde{I}_{\alpha,\beta,k} = I_{\alpha,\beta,k} - \hat{I}_{\alpha,\beta,k} \tag{10}$$

and their dynamics are

$$\tilde{\Phi}_{\alpha,\beta,k+1} = \alpha_d e^{\mathcal{J}hp\omega_k}\tilde{\Phi}_{\alpha,\beta,k} + k_v V_k \tag{11}$$
$$\begin{aligned}\tilde{I}_{\alpha,\beta,k+1} &= \gamma_d\tilde{I}_{\alpha,\beta,k} + \beta_d\alpha_d\tilde{\Phi}_{\alpha,\beta,k} \\ &\quad - \beta_d\alpha_d^2 e^{\mathcal{J}hp\omega_k}\tilde{\Phi}_{\alpha,\beta,k} + V_k.\end{aligned} \tag{12}$$

The sliding function is chosen as $\tilde{I}_{\alpha,\beta,k}$, and V_k as the discrete–time super twisting sliding mode action (Salgado et al. (2011))

$$V_k = -M_3\Gamma_1 \mathrm{sign}(\tilde{I}_{\alpha,\beta,k}) + V_{1,k}$$
$$V_{1,k} = V_{1,k} - h\Big[M_4 \mathrm{sign}(\tilde{I}_{\alpha,\beta,k}) + q_1 V_{1,k}\Big]$$

where $\Gamma_1 = \mathrm{diag}(|\tilde{i}_{\alpha,k}|^{1/2}, |\tilde{i}_{\beta,k}|^{1/2})$, and $\mathrm{sign}(\tilde{I}_{\alpha,\beta,k}) = (\mathrm{sign}(\tilde{i}_{\alpha,k}), \mathrm{sign}(\tilde{i}_{\beta,k}))^T$. The coefficients M_3, M_4 and q_1 are chosen to ensure the sliding mode, as in Salgado et al. (2011). This implies that the current error (10) is zero in a finite–time $k_s > 0$. The corresponding sliding mode dynamics are

$$\tilde{\Phi}_{\alpha,\beta,k+1} = \alpha_d e^{\mathcal{J}hp\omega_k}\tilde{\Phi}_{\alpha,\beta,k} + k_v V_{eq,k} \tag{13}$$

where $V_{eq,k}$ is determined from (12) when $\tilde{I}_{\alpha,\beta,k+1} = 0$

$$V_{eq,k} = -\beta_d\alpha_d\tilde{\Phi}_{\alpha,\beta,k} + \beta_d\alpha_d^2 e^{\mathcal{J}hp\omega_k}\tilde{\Phi}_{\alpha,\beta,k}. \tag{14}$$

The stability analysis of (13) is presented in Rivera (2015b), where it is ensured that (9) tends to zero asymptotically. Once the rotor fluxes are estimated, they are converted, along with the stator currents, to variables in the (d, q) reference frame (2).

3.3 Load torque observer

The load torque can be estimated assuming that it varies slowly with respect to the other dynamics. Therefore, it will be considered a constant signal, i. e., $\tau_{L,k+1} = \tau_{L,k}$. Hence, the following observer is used

$$\begin{aligned}\hat{\omega}_{k+1} &= \hat{\omega}_k - \frac{f_v}{J}\hat{\omega}_k + \mu_d h I^T_{\alpha,\beta,k}\mathcal{J}\hat{\Phi}_{\alpha,\beta,k} \\ &\quad - \frac{h}{J}\hat{\tau}_{L,k} + l_1(\omega_k - \hat{\omega}_k) \\ \hat{\tau}_{L,k+1} &= \hat{\tau}_{L,k} + l_2(\omega_k - \hat{\omega}_k)\end{aligned}$$

with $\hat{\omega}_k$, $\hat{\tau}_{L,k}$ the estimates for the rotor velocity and the load torque, whereas l_1 and l_2 are constant design parameters. The estimation errors are

$$\tilde{\omega}_k = \omega_k - \hat{\omega}_k$$
$$\tilde{\tau}_{L,k} = \tau_{L,k} - \hat{\tau}_{L,k}$$

whose dynamics are

$$\begin{aligned}\tilde{\omega}_{k+1} &= \tilde{\omega}_k - \frac{f_v}{J}\tilde{\omega}_k + \mu_d h I^T_{\alpha,\beta,k}\mathcal{J}\tilde{\Phi}_{\alpha,\beta,k} \\ &\quad - \frac{h}{J}\tilde{\tau}_{L,k} + l_1\tilde{\omega}_k \\ \tilde{\tau}_{L,k+1} &= \tilde{\tau}_{L,k} + l_2\tilde{\omega}_k\end{aligned}$$

or, in matrix form,

$$\Upsilon_{k+1} = A\Upsilon_k + D\tilde{\Phi}_{\alpha,\beta,k} \tag{15}$$

where $\Upsilon = (\tilde{\omega}_k, \tilde{\tau}_{L,K})^T$, and

$$A = \begin{pmatrix} 1 - \frac{f_v}{J} - l_1 & -\frac{h}{J} \\ -l_2 & 1 \end{pmatrix}, \quad D = \begin{pmatrix} \mu_d h I^T_{\alpha,\beta,k}\mathcal{J} \\ 0 \end{pmatrix}.$$

Since $\tilde{\Phi}_{\alpha,\beta,k} = 0$ for $k \geq k_s$, (15) reduces to

$$\Upsilon_{k+1} = A\Upsilon_k, \qquad k \geq k_s. \tag{16}$$

The characteristic equation of (16) is

$$\det(zI - A) = z^2 - (l_1 - 2)z + \left(1 - \frac{f_v}{J} - l_1 - l_2\frac{h}{J}\right). \tag{17}$$

In order to determine values for l_1 and l_2, the coefficients of the polynomial (17) are compared with a desired characteristic polynomial with poles p_1 and p_2, both inside the unit circle in the complex z–plane. Then, l_1 and l_2 are solved, resulting in:

$$l_1 = 2 - p_1 - p_2,$$
$$l_2 = -\frac{h}{J}\left(p_1p_2 - p_1 - p_2 + 1 + \frac{f_v}{J}\right).$$

With this choice, one gets $\lim_{k\to\infty} \Upsilon_k = 0$.

4. SIMULATION RESULTS

The simulations have been conducted considering the values of Table 1 for the IM, and of Table 2 for the controller and the observer.

The rotor velocity reference is zero until $t = 1$ s, when a step of 180 rad/s is introduced. This value remains constant for 1 s. At $t = 2$ s, a step decrements the reference to 100 rad/s, remaining constant for 1 s. Then at $t = 3$ s a ramp decrement is introduced until $t = 4$ s, where a value of 0 rad/s is reached. This value is hold until $t = 5$ s, when a ramp decrement is introduce for 1 s up to -100 rad/s. The reference value is hold for 1 s, and at $t = 7$ s another decrement step is introduced up to -180 rad/s. This value is kept for 1 s until $t = 8$ s, when the reference becomes 0 rad/s. The initial value for the unknown load torque is

Symbol	Value
R_s	$14\,\Omega$
R_r	$10.1\,\Omega$
L_s	$0.4\,\mathrm{H}$
L_r	$0.4128\,\mathrm{H}$
L_m	$0.377\,\mathrm{H}$
L_m	$0.377\,\mathrm{H}$
J	$0.01\,\mathrm{Kg\,m^2}$
p	2

Table 1. IM parameters.

Symbol	Value
$k_{11,p}$	0.94
$k_{12,p}$	0.1
$k_{11,i}$	$-0.001\,1/\mathrm{s}$
$k_{12,i}$	$-0.05\,1/\mathrm{s}$
$k_{21,p}$	0.001
$k_{22,p}$	0.01
$k_{21,i}$	$-15\,1/\mathrm{s}$
$k_{22,i}$	$-20\,1/\mathrm{s}$
k_v	$-0.02\,\mathrm{H}$
M_3	$0.3\sigma/h\,\mathrm{A}^{1/2}$
M_4	$0.045\sigma/h\,\mathrm{As}$
q_1	10
p_1	0.9
p_2	0.8

Table 2. Controller and the observer parameters in the case of the sampling time $h = 100\ \mu$s.

0.5 Nm until $t = 7$ s, when a step increments the torque value to 1.1 Nm. The reference signal for the rotor flux magnitude is the nominal value of 0.4 Wb.

As far as the sampling time value is concerned, in some works available in the literature the sampling time is chosen relatively small, e.g. 50 μs in Gdaim et al. (2015) and 150 μs in Guzinski and Abu-Rub (2013).

Fig. 1 shows the output tracking errors (4) for the proposed control algorithm, for $h = 100\ \mu$s. In the same figure it is shown also the behaviour for an emulated continuous controller, i.e. a continuous controller updated at the sampling instants (see Fekih and Chowdhury (2004) for the design of the continuous controller). It can be appreciated that the angular velocity tracking errors for the two controllers are similar, whereas the rotor flux tracking error is grater for the emulated continuous controller.

To study the controller robustness against parameter variations, and considering that the resistances are the most critical ones, R_s and R_r are varied of +20% in the IM motor, but considered the same as in Table 1 for the proposed control algorithm and the emulated continuous controller. The results are shown in Fig. 2 for $h = 100\ \mu$s.

In order to show that the proposed digital controller gives good results also for (relatively) high sampling periods, we have increased the sampling period h. The emulated controller is unable to give good performance at high sampling periods. Indeed, as shown in Fig. 3 in the case of no parameter variations, and in Fig. 4 for the case of parameter variations, when the sampling time is 150 μs the emulated continuous controller fails to track the velocity

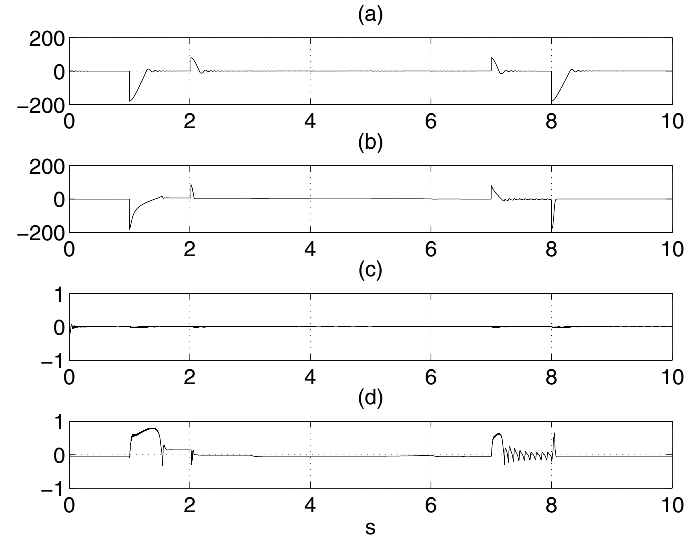

Fig. 1. Output tracking errors without variation parameters with $h = 100\ \mu$s. (a) Rotor velocity error $\omega_k - \omega_{r,k}$ with the proposed controller [rad/s vs s]. (b) Rotor velocity error $\omega - \omega_r$ with the emulated continuous controller [rad/s vs s]. (c) Rotor flux error $\phi_{d,k} - \phi_{d,r,k}$ with the proposed control [Wb vs s]. (d) Rotor flux error $\phi_d - \phi_{d,r}$ with the emulated continuous controller [Wb vs s].

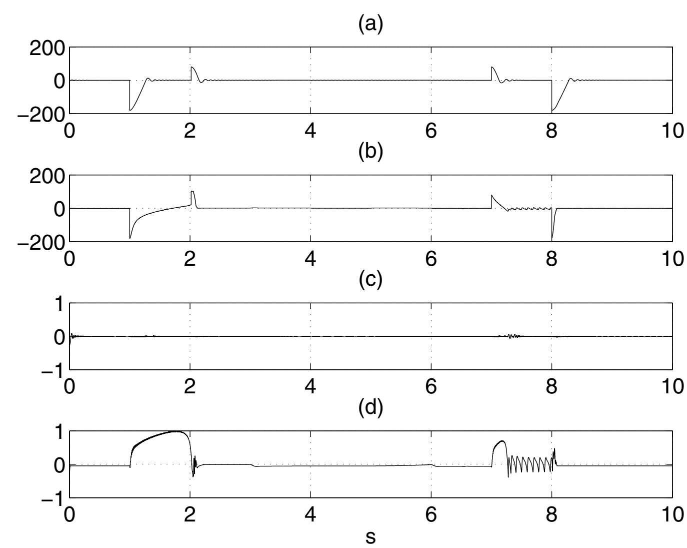

Fig. 2. Output tracking errors with variation parameters with $h = 100\ \mu$s. (a) Rotor velocity error $\omega_k - \omega_{r,k}$ with the proposed controller [rad/s vs s]. (b) Rotor velocity error $\omega - \omega_r$ with the emulated continuous controller [rad/s vs s]. (c) Rotor flux error $\phi_{d,k} - \phi_{d,r,k}$ with the proposed control [Wb vs s]. (d) Rotor flux error $\phi_d - \phi_{d,r}$ with the emulated continuous controller [Wb vs s].

and flux references. On the contrary, the proposed digital controller gives good tracking behaviors. Indeed, analogous behaviors can be noted even for higher sampling periods. For instance, if h is increased to 500 μs, Fig. 5 shows a good tracking of the reference signals, even in the case of higher resistance variations ($R_s = 17.5$, $R_r = 13.2$

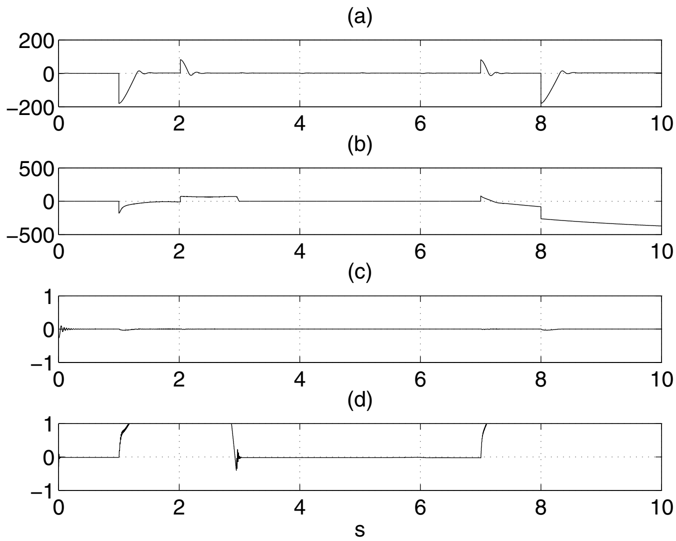

Fig. 3. Output tracking errors without variation parameters with $h = 150$ μs. (a) Rotor velocity error $\omega_k - \omega_{r,k}$ with the proposed controller [rad/s vs s]. (b) Rotor velocity error $\omega - \omega_r$ with the emulated continuous controller [rad/s vs s]. (c) Rotor flux error $\phi_{d,k} - \phi_{d,r,k}$ with the proposed control [Wb vs s]. (d) Rotor flux error $\phi_d - \phi_{d,r}$ with the emulated continuous controller [Wb vs s].

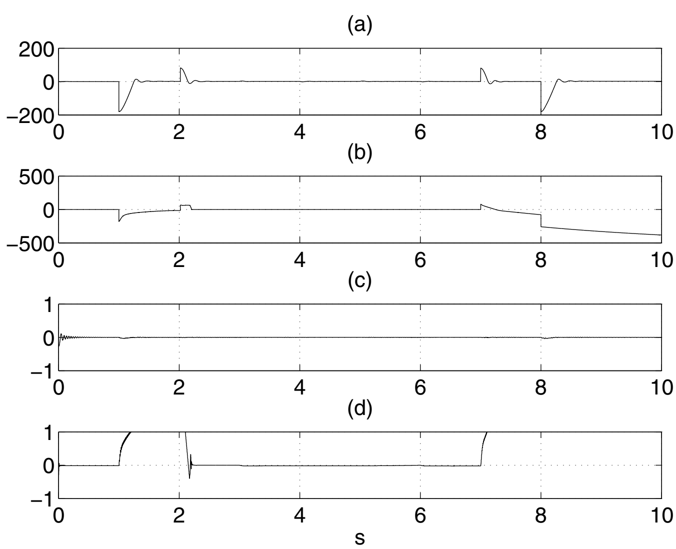

Fig. 4. Output tracking errors with variation parameters with $h = 150$ μs. (a) Rotor velocity error $\omega_k - \omega_{r,k}$ with the proposed controller [rad/s vs s]. (b) Rotor velocity error $\omega - \omega_r$ with the emulated continuous controller [rad/s vs s]. (c) Rotor flux error $\phi_{d,k} - \phi_{d,r,k}$ with the proposed control [Wb vs s]. (d) Rotor flux error $\phi_d - \phi_{d,r}$ with the emulated continuous controller [Wb vs s].

Ω). It has to be noted that in this case the control gains have to be better tuned ($k_{11,p} = 0.97$, $k_{11,i} = -0.002$). The estimation errors from the flux and load observer are shown in Fig. 6, whereas the currents and voltages for the proposed algorithm are shown in Fig. 7.

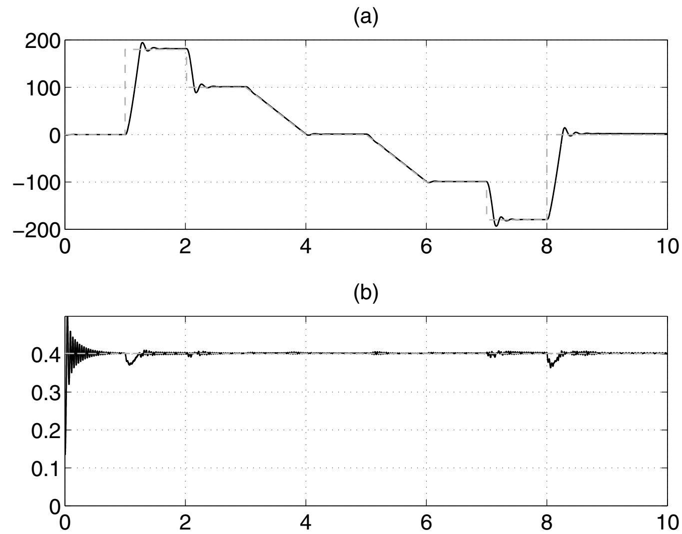

Fig. 5. Sampling period $h = 500$ μs. (a) Rotor velocity ω_k (solid black), and reference signal $\omega_{r,k}$ (dashed gray) [rad/s vs s]. (b) Rotor flux magnitude $\phi_{d,k}$ (solid black), and reference signal $\phi_{d,r,k}$ (dashed gray) [Wb vs s].

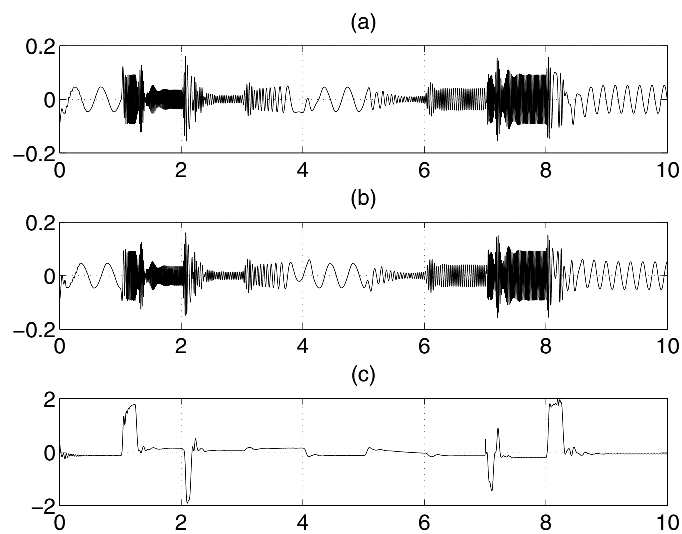

Fig. 6. Sampling period $h = 500$ μs. (a) Rotor flux estimation error $\tilde{\phi}_{\alpha,k}$ [Wb vs s]. (b) Rotor flux estimation error $\tilde{\phi}_{\beta,k}$ [Wb vs s]. (c) Load torque estimation error $\tilde{\tau}_{L,k}$ [Nm vs s].

5. CONCLUSIONS

IMs are often used on industrial applications, and various controllers have been proposed in the literature. Many of them are designed in continuous time, but are implemented on digital devices, so degrading their performance. In many case, these control laws are approximated with an Euler method, using sufficiently small sampling periods to obtain a good performance.

In this work, a recent discrete–time model in the stationary reference frame for IMs is converted to the reference frame oriented to the field, also resulting in a novel model.

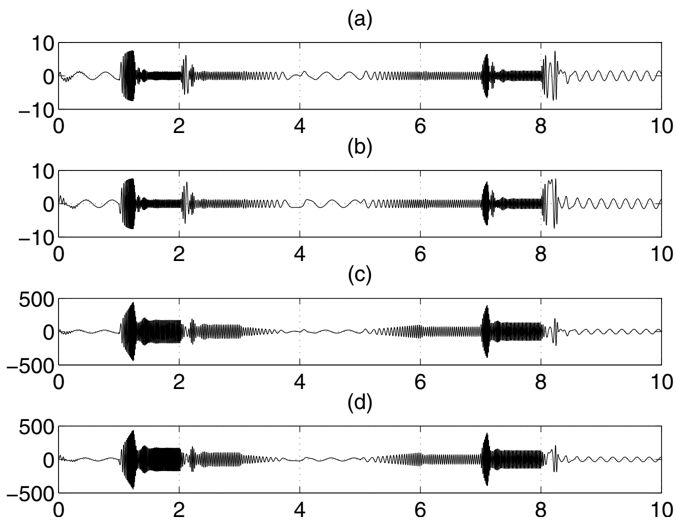

Fig. 7. Sampling period $h = 500$ μs. (a) Rotor Current $I_{\alpha,k}$ [A vs s]. (b) Rotor current $I_{\beta,k}$ [A vs s]. (c) Rotor Voltage U_α [V vs s]. (d) Rotor VoltageU_β [V vs s].

The basic principles of the FOC can be applied on this resulting discrete–time model, i.e. to control each channel individually, with a PI control loop. Hence, a direct discrete–time FOC has been designed in this paper for controlling the rotor velocity and the rotor flux magnitude. For this purpose, a rotor flux observer has been taken from the literature, and a load torque observer has been designed, both in the stationary reference frame.

The simulation results show that the outputs track the reference very accurately. Also, the flux and load torque estimations are very close to the corresponding real signals. Experimental results are contemplated as future work. Some issues still remain, e.g. how to render robust the FOC design, also in the presence of unknown parameter variations.

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